

PRERNA TRIPATHI



8174070826



prerna.tripathi.ece24@itbhu.ac.in



[LinkedIn](#)



[GitHub](#)

Academic Profile

Degree / Certificate	Institution	Percentage / CGPA	Year
B.Tech	Electronics Engineering, IIT (BHU), Varanasi	9.84	2028
CBSE (XII)	Sunbeam English School, Bhagwanpur	96.80	2024
CBSE (X)	Sunbeam English School, Bhagwanpur	97.20	2022

Technical Skills

Languages and Frameworks: C, C++, Python, PyTorch, HTML/CSS

Core Competencies: Competitive Programming, Data Structures and Algorithms, Machine Learning, Deep Learning

Soft Skills: Leadership, Problem Solving, Clear Communication, Confidence, Teamwork, Time Management

Projects

Neural Network for Medical Appointment No-Show Prediction | Python, PyTorch, Deep Learning

- Developed a **PyTorch neural network** for predicting medical appointment **no-shows** using **class-weighted cross-entropy** to handle **class imbalance**, improving minority-class (**No-show**) **recall to 0.77** on an imbalanced dataset.
- Optimized the **decision threshold (0.35)** using **precision–recall curve analysis**, improving the minority-class **F1-score to 0.42** for the clinically critical No-show category.
- Engineered a complete **preprocessing and training pipeline** (**date-feature engineering, one-hot encoding, normalization, balanced DataLoaders, CUDA-accelerated training**), ensuring **stable convergence** across **2700 epochs**.

Weather App | HTML, CSS, JavaScript

- Built a **fully responsive** Weather App allowing users to enter any city and fetch **real-time weather data** via **REST API calls**, ensuring smooth performance across devices.
- Implemented **client-side JavaScript logic** to process **API responses** and dynamically render **temperature, wind speed, date, and condition icons**, enabling **live updates** without page refresh.
- Enhanced **user experience (UX)** by integrating **Animate.css** for **fluid transitions** and interactive visual cues, making weather changes more intuitive and engaging.

Positions of Responsibility

Electronics Engineering Society, IIT BHU

Oct 2025 – Present

Technical Co-Coordinator of SyNNapse

- Leading the technical planning and end-to-end execution of **SyNNapse**, a deep learning event at **Udyam'26**, including coordination, problem design, and event logistics.

Club of Programmers, IIT BHU

July 2025 – Present

Member of the Intelligence Guild

- Conducted technical orientations and workshops under the COPS Intelligence Guild, focusing on programming fundamentals, problem-solving, and collaborative learning.

Honours and Achievements

- Achieved a **Highest Rating of 1235** on [Codeforces](#).
- Secured **3rd Position** in the Confluentia Hackathon held by SAIC, IIT BHU, involving a deep learning model for query–category relevance.
- Secured **3rd Position** in Funkit'25, a digital electronics event at Udyam'25.
- Solved **700+ problems** on Codeforces and LeetCode.
- Secured **AIR 2626** in **JEE Mains 2024** and **AIR 5767** in **JEE Advanced 2024**.

Extra-Curricular Activities

- Secured **1st Position** in Storytelling Competition at **Aagman'24**, an intrafreshers event of IIT BHU.